

WILBARGER COUNTY AP, TX, USA

WMO: 720281

Lat: 34.226N

Lon: 99.284W

Elev: 1265

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	23.0	0.8	6.0	31.6	5.7	7.7	33.7	29.7	47.3	26.6	49.5	8.0	20	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	105.6	70.7	102.3	70.9	99.8	70.7	76.2	91.2	75.2	90.2	74.3	89.4	11.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	127.5	80.4	71.9	123.6	80.1	70.3	117.0	80.0	40.5	90.6	39.4	89.8	38.6	88.4	95.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	23.8	20.5	DB	12.1	107.7	5.2	3.6	8.4	110.3	5.4	112.4	2.5	114.5	-1.3	117.1	
			WB	10.5	77.3	4.9	1.8	6.9	78.5	4.0	79.6	1.3	80.6	-2.3	81.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	43.0	46.8	56.2	63.5	73.3	82.8	85.6	85.0	78.1	65.6	54.5	43.9
	DBStd	17.45	9.26	10.70	10.09	8.83	8.43	5.31	4.94	5.52	6.36	8.97	9.02	9.18
	HDD50	775	253	169	52	9	0	0	0	0	0	7	56	229
	HDD65	2742	682	515	303	129	27	0	0	0	1	104	326	655
	CDD50	6238	36	79	246	414	722	985	1105	1086	843	490	192	40
	CDD65	2730	0	5	31	84	284	535	640	621	395	122	12	1
	CDH74	35473	26	106	415	1124	3327	6970	9003	8437	4509	1334	200	22
	CDH80	19758	3	26	117	441	1651	4030	5495	5107	2330	511	44	3
(14) Wind	WSAvg	9.7	9.4	10.1	11.1	11.7	10.6	10.7	8.8	8.3	8.3	9.3	9.5	9.0
(15) Precipitation	PrecAvg	27.3	1.1	1.4	1.9	2.6	3.4	4.0	2.4	2.4	2.5	2.6	1.6	1.2
	PrecMax	42.4	2.5	4.1	3.8	6.4	7.6	10.7	7.7	5.4	6.4	8.5	7.2	4.5
	PrecMin	16.3	0.0	0.0	0.0	0.1	0.6	0.1	0.2	0.0	0.3	0.0	0.0	0.0
	PrecStd	7.2	0.8	1.1	1.2	1.8	1.9	2.8	1.9	1.6	1.8	2.1	1.8	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.5	83.2	89.1	96.1	101.8	107.9	108.4	109.1	102.1	93.4	85.8	77.0
		MCWB	52.1	57.5	59.9	64.2	65.9	70.1	70.7	70.7	69.5	67.2	61.4	56.4
	2%	DB	70.5	76.6	82.8	90.1	97.8	102.2	104.4	105.6	98.4	89.9	79.5	69.8
		MCWB	50.5	55.6	60.3	63.9	68.2	71.2	71.1	71.2	69.4	66.9	58.8	51.9
	5%	DB	64.8	70.6	78.9	85.1	93.6	99.2	101.9	102.3	95.2	86.1	75.0	63.9
		MCWB	49.7	53.0	58.9	62.6	68.4	71.0	71.0	71.4	69.8	64.8	57.8	50.5
	10%	DB	59.9	64.7	73.5	80.7	89.6	96.6	99.3	99.4	91.7	81.7	70.2	59.4
		MCWB	46.2	51.6	57.2	61.7	67.5	71.4	71.1	71.6	69.1	63.1	56.3	49.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	62.2	67.5	73.3	75.3	77.6	79.2	77.2	76.0	73.1	65.8	61.4
		MCDB	67.9	71.2	77.5	86.1	88.6	93.4	93.6	93.6	89.2	84.2	77.9	69.1
	2%	WB	54.0	59.1	65.4	69.0	73.6	75.6	76.1	75.6	74.5	70.9	62.9	56.8
		MCDB	64.0	70.2	75.2	80.2	88.4	91.5	90.1	91.5	86.4	82.3	72.4	64.6
	5%	WB	50.4	55.6	63.1	66.8	72.3	74.5	74.8	74.7	73.3	68.9	60.8	52.4
		MCDB	62.5	68.2	72.4	78.2	86.1	90.8	89.3	90.8	84.5	79.9	70.9	62.0
	10%	WB	47.2	52.0	60.0	64.7	70.7	73.3	73.6	73.6	72.1	66.7	58.4	48.7
		MCDB	59.8	65.0	70.2	75.6	83.8	89.9	88.6	90.1	84.1	77.4	68.1	59.4
(35) Mean Daily Temperature Range	5% DB	MDBR	24.7	24.3	25.8	26.0	24.1	23.2	23.6	23.6	23.2	24.7	24.6	22.6
		MCDBR	34.5	34.8	32.2	33.1	29.6	27.1	27.5	27.5	26.8	29.2	31.2	31.2
		MCWBR	18.8	17.4	15.1	14.8	9.6	7.5	6.4	6.6	6.9	11.2	15.3	17.6
		MCDBR	31.3	29.7	23.9	26.4	24.3	23.3	22.0	23.2	20.5	21.9	25.5	27.2
	5% WB	MCWBR	18.8	17.6	12.9	14.6	9.9	8.0	6.9	6.0	6.4	10.6	14.8	17.6
(40) Clear-Sky Solar Irradiance	taub		0.286	0.299	0.328	0.364	0.390	0.418	0.417	0.410	0.379	0.348	0.312	0.288
	taud		2.535	2.506	2.442	2.359	2.358	2.315	2.365	2.328	2.391	2.462	2.495	2.529
	Ebn at Noon		297	303	301	292	284	275	274	275	279	281	283	289
	Edh at Noon		26	30	34	39	39	41	39	40	36	30	27	24
(44) All-Sky Solar Radiation	RadAvg		973	1164	1553	1885	2025	2228	2217	2023	1684	1307	1015	831
	RadStd		98	160	98	147	208	185	129	119	116	178	126	85

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page